Poisson–Dirichlet Universality for Luce Permutations

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Abstract

Let a Luce permutation be generated by ranking independent exponential clocks with prescribed positive rates. Under a uniform bound on the ratio of the largest to the smallest rate, the normalized ranked cycle lengths converge in ℓ^1 to PD(1), and their size-biased deletion order converges in ℓ^1 to GEM(1). In particular, the cycle length of a uniformly chosen label, divided by the number of labels, converges to the uniform distribution on $(0, 1)$.

This manuscript was mechanically extracted from the macroscopic-cycle section of the archived 139-page Luce manuscript at pinned repository commit `6178a99`. A subsequent adversarial audit found no fatal gap in the stated theorem. The present revision applies only the six localized clarifications recorded in the accompanying audit and repair records; it does not change a principal hypothesis, conclusion, topology, or proof strategy. The review is internal and is not formal verification or independent external refereeing.

1 Model, scope, and proof dependencies

Let $X_{n,i} \sim \mathrm{Exp}(\theta_{n,i})$, $1 \le i \le n$, be independent clocks with positive rates. The label-to-position Luce permutation is

$$\sigma_n(i) = 1 + \#\{j : X_{n,j} < X_{n,i}\}.$$

Multiplying all rates by a common positive constant does not change the permutation.

All model-specific ingredients used in the macroscopic-cycle proof are stated and proved below: the subsequential marked kernel and its contraction, the clock-slab and residual permanental structure, hereditary flatness, the relative-prefix estimate, two-scale orbit balance, the rooted closure law, delete-and-regenerate stability, and the finite-dimensional GEM induction. The proof does not import the mesoscopic-cycle theorem, the total-cycle CLT, or any result from another active manuscript in this repository.

The external inputs are standard. For weak compactness and disintegration on standard Borel spaces we use Kallenberg [4, Chapters 4 and 6]; for Kolmogorov's maximal inequality and Doob's L^2 maximal inequality we use Durrett [3, Chapters 2 and 4]; and for convergence of expectations against bounded continuous functions and the continuous mapping theorem we use Billingsley [1, Theorems 2.1 and 2.7]. The bounded-Lipschitz criterion on separable metric spaces is recorded in Dudley [2, Section 11.3]. The standard relation between GEM(1) and PD(1) is cited to Pitman [5, Chapter 3]. The old-label/new-label register accompanying this extraction records every theorem, proposition, lemma, equation label, and bibliographic key moved from the archive.

2 Macroscopic cycles under uniformly comparable weights

The preceding results concern cycles whose lengths remain fixed as $n \to \infty$. We now consider cycles of order n. Under a uniform bound on the ratio of the largest to the smallest weight, the macroscopic cycle partition is universal. We write GEM(1) for the residual-allocation sequence

$$U_1, \quad (1-U_1)U_2, \quad (1-U_1)(1-U_2)U_3, \ldots,$$

where the U_j are independent uniform random variables, and PD(1) for its decreasing rearrangement; see Pitman [5, Chapter 3]. Let

$$\mathcal{S}^{\downarrow} = \left\{ x_1 \geq x_2 \geq \cdots \geq 0 : \sum_{j \geq 1} x_j = 1 \right\},$$

equipped with the ℓ^1 metric.

Theorem 2.1 (Macroscopic cycle universality)**.** *Let σ_n be a Luce permutation with positive weights $\theta_{n,1}, \ldots, \theta_{n,n}$ and assume*

$$\sup_n \frac{\max_i \theta_{n,i}}{\min_i \theta_{n,i}} < \infty.$$

If $L_{n,1} \geq L_{n,2} \geq \cdots$ are the cycle lengths of σ_n, then

$$\left(\frac{L_{n,1}}{n}, \frac{L_{n,2}}{n}, \ldots \right) \Rightarrow \mathrm{PD}(1)$$

in $\mathcal{S}^{\downarrow}$. Moreover, if $\widetilde{L}_{n,1}, \widetilde{L}_{n,2}, \ldots$ are obtained by choosing, at each stage, a new vertex independently and uniformly from the current residual vertex set, listing the length of its cycle, deleting that cycle, and repeating, then

$$\left(\frac{\widetilde{L}_{n,1}}{n}, \frac{\widetilde{L}_{n,2}}{n}, \ldots \right) \Rightarrow \mathrm{GEM}(1)$$

in ℓ^1. In particular, if I_n is a uniformly chosen label and $\mathscr{L}_n(I_n)$ is the length of its cycle, then

$$\frac{\mathscr{L}_n(I_n)}{n} \Rightarrow \mathrm{Unif}(0,1).$$

The proof uses one fixed finite partition of clock time. Conditional on the clock slabs, each slab row is a permanental matching whose entries are nearly constant. A common minorization gives a spectral gap depending only on the weight-comparability constant. A relative-prefix theorem then controls every target category under arbitrary adaptive exploration. We formulate all concentration statements uniformly over an explicit class of regular residual histories, and prove that deleting an interior macroscopic cycle preserves that class. This supplies the regeneration needed for the GEM(1) induction.

Multiplying all weights by a common positive factor does not change the Luce order. We therefore normalize throughout the proof so that, for a fixed $\kappa < \infty$,

$$1 \leq \theta_{n,i} \leq \kappa, \qquad 1 \leq i \leq n. \tag{6.0}$$

2.1 Subsequential marked kernels and a uniform contraction

Fix an arbitrary subsequence. Passing to a further subsequence, compactness gives

$$\eta_n := \frac{1}{n}\sum_{i=1}^{n} \delta_{(i/n,\theta_{n,i})} \Longrightarrow \eta$$

on $[0,1]\times[1,\kappa]$. The first marginal of η is Lebesgue measure. Choose a disintegration

$$\eta(\,dx,\,d\vartheta) = \,dx\,\eta_x(\,d\vartheta),$$

defined for almost every x. Put

$$A(t) = \int (1-e^{-\vartheta t})\,\eta(\,dx,\,d\vartheta), \qquad B(t) = A'(t) = \int \vartheta e^{-\vartheta t}\,\eta(\,dx,\,d\vartheta),$$

let $\tau = A^{-1}$, and define, for almost every x,

$$g_x(t) = \int \vartheta e^{-\vartheta t}\,\eta_x(\,d\vartheta), \qquad \rho(x,y) = \frac{g_x(\tau(y))}{B(\tau(y))}.$$

Let K be the associated Markov operator on $L^2[0,1]$.

Lemma 2.2 (Bistochasticity and common minorization)**.** *The kernel ρ is bistochastic up to null sets and*

$$\iint_{[0,1]^2} \rho(x,y)^2\,dx\,dy \le \kappa.$$

On the mean-zero subspace,

$$\|K\|_{L^2_0\to L^2_0} \le q_\kappa := \sqrt{1-\kappa^{-1}} < 1,$$

and the same bound holds for K^*.

Proof. Using $y = A(t)$ and $dy = B(t)\,dt$,

$$\int_0^1 \rho(x,y)\,dy = \int_0^\infty g_x(t)\,dt = 1$$

for almost every x, while

$$\int_0^1 \rho(x,y)\,dx = \frac{\int_0^1 g_x(\tau(y))\,dx}{B(\tau(y))} = 1.$$

Jensen's inequality and $\vartheta \le \kappa$ give

$$\int_0^1 g_x(t)^2\,dx \le \int \vartheta^2 e^{-2\vartheta t}\,\eta(\,dx,\,d\vartheta) \le \kappa B(2t).$$

Since $\vartheta \ge 1$, $B(2t) \le e^{-t}B(t)$, and hence

$$\iint \rho^2 = \int_0^\infty \frac{\int g_x(t)^2\,dx}{B(t)}\,dt \le \kappa.$$

For the contraction, regard $g_x(t)\,dt$ as the conditional law of a clock time T given x. From (6.0), $g_x(t) \geq e^{-\kappa t}$. Let

$$\alpha = \kappa^{-1}, \qquad \nu(\,dt) = \kappa e^{-\kappa t}\,dt.$$

There are probability kernels R_x such that

$$g_x(t)\,dt = \alpha\nu(\,dt) + (1-\alpha)R_x(\,dt).$$

The marginal law of T is $\mu = \alpha\nu + (1-\alpha)\bar{R}$, where $\bar{R} = \int_0^1 R_x\,dx$. If $f \in L^2(\mu)$ and $\mu(f) = 0$, then

$$\int f(t)g_x(t)\,dt = (1-\alpha)\{R_x(f) - \bar{R}(f)\}.$$

Jensen's inequality therefore yields

$$\begin{aligned}\int_0^1 \left|\int f(t)g_x(t)\,dt\right|^2 dx &\leq (1-\alpha)^2\bar{R}(f^2)\\ &\leq (1-\alpha)\mu(f^2).\end{aligned}$$

The change of variables $y = A(t)$ is an isometry from $L^2(\,dy)$ to $L^2(\mu)$, and under this isometry the displayed conditional-expectation operator is K. This proves the bound; the adjoint has the same norm. □

2.2 Clock slabs and residual permanental matchings

Fix $T > 0$ and a partition

$$0 = t_0 < t_1 < \cdots < t_d = T, \qquad \max_a(t_a - t_{a-1}) \leq h,$$

and append the tail interval $I_\star = (T, \infty)$. Put

$$I_a = (t_{a-1}, t_a] \quad (a \leq d), \qquad H_{n,a} = \{i : X_{n,i} \in I_a\}, \qquad m_{n,a} = |H_{n,a}|.$$

If $M_{n,a} = \sum_{c\leq a} m_{n,c}$, then the ranks occupied by $H_{n,a}$ are

$$Q_{n,a} = \{M_{n,a-1} + 1, \ldots, M_{n,a}\};$$

the tail row is defined analogously. Set

$$N_{n,ab} = |Q_{n,a} \cap H_{n,b}|, \qquad p_{n,a} = m_{n,a}/n.$$

When $m_{n,a} > 0$, put $P_{n,ab} = N_{n,ab}/m_{n,a}$; if $m_{n,a} = 0$, define that row of P_n arbitrarily. On the good events used below every row is nonempty. Thus P_n is stochastic and $p_nP_n = p_n$.

Let

$$J_a = (A(t_{a-1}), A(t_a)] \quad (a \leq d), \qquad J_\star = (A(T), 1], \qquad p_a = |J_a|,$$

and define

$$q_{ab} = \int_{J_a\times J_b} \rho(x,y)\,dx\,dy, \qquad P_{ab} = q_{ab}/p_a.$$

Every p_a and q_{ab} is strictly positive.

Lemma 2.3 (Slab law and deterministic good events)**.** *For every fixed partition,*

$$p_{n,a} \xrightarrow{\mathbb{P}} p_a, \qquad N_{n,ab}/n \xrightarrow{\mathbb{P}} q_{ab}$$

simultaneously in a, b. There are deterministic $\varepsilon_n \downarrow 0$ and events $\mathcal{G}_n$ with $\mathbb{P}(\mathcal{G}_n) \to 1$ such that, on $\mathcal{G}_n$:

(i) all of the preceding convergences hold with error at most ε_n^2, and, for all sufficiently large n,

$$p_{n,a} \geq \tfrac{1}{2} p_a, \qquad N_{n,ab} \geq \tfrac{1}{2} q_{ab} n$$

for every a, b;

(ii) for a constant $q_0 \in (q_\kappa, 1)$,

$$\|P_n\|_{L_0^2(p_n) \to L_0^2(p_n)} \leq q_0;$$

(iii) if S is a partition endpoint and $J_b \subset [0, A(S)]$, then

$$P_{n,ab} \leq 2 M_S p_{n,b} \qquad \text{for every } a, \qquad M_S = \kappa e^{(\kappa-1)S}; \tag{6.1}$$

(iv) for every union D of slab states,

$$\sum_{a \in D} \sum_b p_{n,a} \frac{P_{n,ab}^2}{p_{n,b}} \leq 2\kappa; \tag{6.2}$$

(v) if U is a partition endpoint and $D(U)$ is the union of rows whose clock intervals lie in (U, ∞), then

$$\sum_{a \in D(U)} p_{n,a} \leq 2e^{-U} + \varepsilon_n^2. \tag{6.3}$$

The constants may depend on the fixed partition and on S, but not on n.

Proof. For a fixed clock cell $I_b = (u, v]$, the variables $\mathbf{1}_{\{X_{n,i} \in I_b\}}$ are independent and have means $e^{-\theta_{n,i} u} - e^{-\theta_{n,i} v}$. Kolmogorov's maximal inequality gives

$$\max_{k \leq n} \left| \sum_{i \leq k} \left(\mathbf{1}_{\{X_{n,i} \in I_b\}} - \mathbb{E}\mathbf{1}_{\{X_{n,i} \in I_b\}} \right) \right| = o_{\mathbb{P}}(n).$$

Marked empirical convergence and the Lebesgue first marginal imply, uniformly in $x \in [0, 1]$,

$$\frac{1}{n} \sum_{i \leq nx} \mathbb{E}\mathbf{1}_{\{X_{n,i} \in I_b\}} \longrightarrow \int_0^x \int (e^{-\vartheta u} - e^{-\vartheta v}) \eta_z(\, d\vartheta)\, dz.$$

This proves the slab-size convergence and convergence of the endpoints of $Q_{n,a}/n$ to those of J_a. Applying the same uniform partial-sum estimate at those random endpoints gives

$$\frac{N_{n,ab}}{n} \longrightarrow \int_{J_a} \int (e^{-\vartheta t_{b-1}} - e^{-\vartheta t_b}) \eta_x(\, d\vartheta)\, dx = q_{ab}.$$

The last equality follows by integrating $\rho(x, y)$ over J_b and using $y = A(t)$.

The matrix P is the compression of K to functions that are constant on the sets J_a: explicitly, it is $\Pi_{\mathcal{J}} K \Pi_{\mathcal{J}}$. Conditional expectation is a contraction, so Lemma 2.2 gives $\|P\|_{L^2_0(p)} \le q_\kappa$. Entrywise convergence of the conjugated matrices $D_{p_n}^{1/2} P_n D_{p_n}^{-1/2}$ proves (ii).

If $y \le A(S)$, then $\tau(y) \le S$ and

$$\rho(x,y) = \frac{g_x(\tau(y))}{B(\tau(y))} \le \frac{\kappa e^{-\tau(y)}}{e^{-\kappa\tau(y)}} \le \kappa e^{(\kappa-1)S} =: M_S.$$

Hence $P_{ab} \le M_S p_b$, which yields (6.1) by entrywise convergence. On every rectangle, Cauchy's inequality gives

$$p_a \frac{P_{ab}^2}{p_b} = \frac{(\int_{J_a \times J_b} \rho)^2}{p_a p_b} \le \int_{J_a \times J_b} \rho^2.$$

Summing and using Lemma 2.2 proves the limiting version of (6.2), and entrywise convergence gives its finite-n form. Finally, the expected proportion of clocks exceeding U is at most e^{-U} because all rates are at least one; the slab law gives (6.3). Since the partition is finite, choose deterministic $\delta_n \downarrow 0$ so that all convergence assertions hold with error at most δ_n on a single event of probability tending to one, and set $\varepsilon_n = \sqrt{\delta_n}$. This gives the stated event $\mathcal{G}_n$ and the errors ε_n^2 in (i) and (6.3). □

Condition on the complete slab data: the sets $H_{n,a}$, the rank intervals $Q_{n,a}$, and the ordered clock values

$$s_{a,1} < \cdots < s_{a,m_{n,a}}$$

in every slab. The ordered list is *unlabeled*: the conditioning sigma-field does not record which label in $H_{n,a}$ realizes which value $s_{a,j}$, and hence it does not determine the ranking inside the slab. Write $Q_{n,a} = \{q_{a,1} < \cdots < q_{a,m_{n,a}}\}$, and let $\phi_a : H_{n,a} \to [m_{n,a}]$ be defined by $\sigma_n(i) = q_{a,\phi_a(i)}$. For a bijection $\phi : H_{n,a} \to [m_{n,a}]$, the joint exponential density of the assignment $i \mapsto s_{a,\phi(i)}$ is

$$\prod_{i \in H_{n,a}} \theta_{n,i} e^{-\theta_{n,i} s_{a,\phi(i)}}.$$

Summing this product over all bijections gives the conditional normalizing constant. The factors $\prod_i \theta_{n,i}$ do not depend on the assignment and cancel, so

$$\mathbb{P}(\phi_a = \phi \mid \text{slab data}) = \frac{\prod_{i \in H_{n,a}} e^{-\theta_{n,i} s_{a,\phi(i)}}}{\operatorname{per}(e^{-\theta_{n,i} s_{a,j}})_{i,j}}. \tag{6.3a}$$

Because the original clock density is a product and the slab label sets are disjoint, these conditional assignment laws factor over the slabs. Thus the row matchings are conditionally independent.

Lemma 2.4 (Residual permanental factorization)**.** *Condition on the slab data and on any compatible partial collection of exposed matching edges. In every row, delete the exposed source and target vertices. The remaining row matchings are conditionally independent, and each has the permanental law* (6.3a) *on its remaining submatrix.*

Proof. The conditional density of the complete matching is a product over rows and, within each row, a product over its edges. Conditioning on fixed exposed edges removes their constant factors and restricts the sum to bijections of the remaining rows and columns. The normalizing sum is the permanent of the remaining submatrix, and the factorization over rows is unchanged. □

Lemma 2.5 (Hereditary flatness)**.** *Let $A = (a_{ij})$ be a positive $m \times m$ matrix satisfying*

$$e^{-\epsilon} \le a_{ij}/a_{ik} \le e^{\epsilon}$$

for every row i and every pair of columns j, k. Under the permanental matching law proportional to $\prod_i a_{i,\phi(i)}$, after conditioning on an arbitrary partial matching, every remaining edge probability satisfies

$$\frac{e^{-2\epsilon}}{s} \le \mathbb{P}(\phi(i) = j \mid \textit{partial matching}) \le \frac{e^{2\epsilon}}{s},$$

where s rows and columns remain. For a Luce slab of width at most h, this holds with $\epsilon = \kappa h$; write

$$\beta = 2\kappa h. \tag{6.4}$$

Proof. It suffices to work with the current remaining submatrix. For two remaining columns j, k,

$$\frac{\mathbb{P}(\phi(i) = j)}{\mathbb{P}(\phi(i) = k)} = \frac{a_{ij} \operatorname{per} A_{-i,-j}}{a_{ik} \operatorname{per} A_{-i,-k}}.$$

Map a term of the first minor to a term of the second by locating the row sent to k and sending that row to j instead. Corresponding products differ by a factor in $[e^{-\epsilon}, e^{\epsilon}]$. Including the factor a_{ij}/a_{ik} shows that every pair of edge probabilities has ratio in $[e^{-2\epsilon}, e^{2\epsilon}]$. Since the s probabilities sum to one, the individual bounds follow. In a clock slab,

$$e^{-\kappa h} \le e^{-\theta_i(s_j - s_k)} \le e^{\kappa h}.$$

□

Lemma 2.6 (Predictable row exposure)**.** *Condition on the slab data and on a compatible partial collection of exposed matching edges. Let $(\mathscr{F}_r)$ be an exploration filtration such that, before the $(r+1)$st edge is exposed, its source is an unexposed source that is $\mathscr{F}_r$-measurable. Then, conditional on $\mathscr{F}_r$, the target of that source has the one-edge marginal of the residual permanental matching in its row. In particular, if that row has s targets remaining and has width at most h, every individual target has conditional probability in*

$$[e^{-\beta}/s, e^{\beta}/s], \qquad \beta = 2\kappa h.$$

If a distinguished target v in that row is forbidden and one conditions the next target not to equal v, then every one of the other $s-1$ targets has conditional probability in

$$[e^{-\beta}/(s-1), e^{\beta}/(s-1)].$$

The conclusions remain true when the next source is selected adaptively using all previously exposed edges, including edges from the same row.

Proof. By Lemma 2.4, conditional on $\mathscr{F}_r$ the unexposed part of the relevant row has the permanental law on its current submatrix. Lemma 2.5 therefore gives the first interval of probabilities. If v is forbidden, conditioning away from v merely renormalizes the probabilities of the other targets. Their pairwise ratios are unchanged and lie in $[e^{-\beta}, e^{\beta}]$; since their conditional probabilities sum to one, each lies in the second displayed interval. The measurability assumption is precisely what is needed to apply the residual-factorization statement before the new edge is revealed. No independence between the chosen source and the previously exposed edges is required. □

2.3 A quantitative relative-prefix theorem

Lemma 2.7 (Relative prefix law)**.** *Fix $d \geq 1$ and $c_* > 0$. There are absolute constants $\beta_0 > 0$ and $C < \infty$, and deterministic sequences $\chi_m \downarrow 0$ and $\pi_m \downarrow 0$, depending only on d and c_*, with the following property. Let m targets be partitioned into d categories of sizes $m_1, \dots, m_d$ satisfying $m_b \geq c_* m$. Targets are selected sequentially without replacement in an arbitrary predictable, adaptive source order. Suppose that, whenever s targets remain, every individual remaining target has conditional selection probability in*

$$[e^{-\beta}/s, e^{\beta}/s], \qquad 0 \leq \beta \leq \beta_0.$$

If $X_b(r)$ is the number of category-b targets among the first r selections, then

$$\mathbb{P}\left(\max_{b \leq d} \max_{0 \leq r \leq m} \frac{|X_b(r) - r m_b/m|}{m_b} > C\beta + \chi_m \right) \leq \pi_m. \tag{6.5}$$

The estimate is uniform over all adaptive source rules satisfying the displayed one-target bounds. It remains valid after deleting any fixed number of distinguished targets before the procedure, or after forbidding those targets throughout the procedure, after changing c_* and the two deterministic sequences harmlessly.

Proof. Fix one category and write

$$N_r = m_b - X_b(r), \qquad s_r = m - r.$$

Let Y_{r+1} be the indicator that the next selected target belongs to this category. By summing the individual-target bounds over the N_r remaining targets in the category,

$$\pi_r := \mathbb{E}(Y_{r+1} \mid \mathscr{F}_r) \in \left[e^{-\beta} \frac{N_r}{s_r}, e^{\beta} \frac{N_r}{s_r} \right]. \tag{6.6}$$

Set $a_m = m_b^{2/3}$ and stop at

$$\tau = \inf\{r : N_r < a_m \text{ or } s_r < a_m\}.$$

For $r < \tau$, put $Z_r = \log(N_r/s_r)$. Since

$$Z_{r+1} - Z_r = -\log(1 - s_r^{-1}) + Y_{r+1} \log(1 - N_r^{-1}),$$

Taylor's formula and (6.6) give

$$|\mathbb{E}(Z_{r+1} - Z_r \mid \mathscr{F}_r)| \leq C_1 \frac{\beta}{s_r} + C_1 \left(\frac{1}{s_r^2} + \frac{1}{s_r N_r} \right), \tag{6.7}$$

and the conditional variance of the centered increment is at most $C_1/(s_r N_r)$. Because $m_b \geq c_* m$,

$$\sum_{r < \tau} \frac{1}{s_r N_r} \leq a_m^{-1} \sum_{s \geq a_m} \frac{1}{s} \leq C_{c_*} \frac{1 + \log m}{m^{2/3}} =: v_m, \tag{6.8}$$

and the sum of the two second-order drift terms in (6.7) is $O(v_m)$.

Let M_r be the stopped martingale obtained by summing the centered increments. Doob's L^2 inequality and (6.8) imply

$$\mathbb{P}\left(\max_{r\le\tau}|M_r| > v_m^{1/4}\right) \le C v_m^{1/2}. \tag{6.9}$$

On the complementary event, for every $r < \tau$,

$$\left|\log\frac{N_r/m_b}{s_r/m}\right| \le C_2\beta\log\frac{m}{s_r} + C_2(v_m^{1/4} + v_m). \tag{6.10}$$

Choose β_0 so that

$$C_2\beta_0 \le \frac{1}{2}, \qquad \beta_0 \le \log 2.$$

The second harmless restriction ensures that every later harmonic-product exponent $e^{\pm\beta}$ lies in $[1/2, 2]$. Writing $u = s_r/m$ and $\eta_m = C_2(v_m^{1/4} + v_m)$, (6.10) yields

$$e^{-\eta_m}u^{1+C_2\beta} \le \frac{N_r}{m_b} \le e^{\eta_m}u^{1-C_2\beta}. \tag{6.11}$$

For $|a| \le 1/2$,

$$\sup_{0\le u\le 1}|u^{1+a} - u| \le C|a|,$$

which follows from the mean-value theorem in the exponent and $\sup_{0<u\le1} u|\log u| < \infty$. Hence, before τ,

$$\left|\frac{N_r}{m_b} - \frac{s_r}{m}\right| \le C_3\beta + C_3\eta_m. \tag{6.12}$$

At the stopping time there are two cases. If $s_\tau < a_m$, then $s_\tau/m = O(m^{-1/3})$. At time $\tau - 1$ we have $s_{\tau-1} < a_m + 1$, and the upper bound in (6.11) gives

$$\frac{N_\tau}{m_b} \le \frac{N_{\tau-1}}{m_b} \le e^{\eta_m}\left(\frac{a_m+1}{m}\right)^{1-C_2\beta} = o(1),$$

uniformly because $1 - C_2\beta \ge 1/2$. If instead $N_\tau < a_m$, then $N_{\tau-1} < a_m + 1$ and the lower bound in (6.11) yields

$$\frac{s_{\tau-1}}{m} \le \left(e^{\eta_m}\frac{a_m+1}{m_b}\right)^{1/(1+C_2\beta)} = o(1),$$

so also $s_\tau/m = o(1)$. Thus in either case both normalized remaining counts are $o(1)$ at τ, up to the already recorded $O(\beta)$ error. After τ each normalized count is nonincreasing, so the absolute difference is bounded by the sum of its two values at τ and hence by $C_3\beta + o(1)$. Therefore (6.12) extends to all $0 \le r \le m$.

For example, one may take

$$\chi_m = C_{d,c_*}\left(v_m^{1/4} + m^{-1/12}\right), \qquad \pi_m = C_{d,c_*}v_m^{1/2}.$$

A union bound over the fixed number of categories proves (6.5). Deleting or forbidding finitely many targets changes every relevant category size and the total size by $O(1)$; for all large m the positive-proportion hypothesis and the same one-target ratio bounds remain valid after changing c_* by a fixed factor. □

In every application below the mesh is chosen sufficiently fine that $\beta = 2\kappa h \le \beta_0$.

2.4 Regular residual histories and scale-sensitive orbit balance

A residual history $\mathcal{H}$ consists of the complete slab data, all roots selected so far, and the matching edges in a collection of fully exposed and deleted cycles; it contains no additional information about unexposed row matchings. After such a history, let $R \subset [n]$ be the remaining vertex set and $N = |R|$. Put

$$m_a^R = |H_{n,a} \cap R|, \qquad N_{ab}^R = |Q_{n,a} \cap H_{n,b} \cap R|, \qquad r = N/n.$$

For every deleted cycle C and every slab a,

$$|C \cap H_{n,a}| = |C \cap Q_{n,a}|, \tag{1}$$

because σ_n maps $C \cap H_{n,a}$ bijectively onto $C \cap Q_{n,a}$. Consequently $m_a^R = |Q_{n,a} \cap R|$.

Let U be a master-partition endpoint. A residual history is called (U, δ, ξ)-regular if $N \geq \xi n$, $0 \leq \delta < 1/4$, and, for every row whose clock interval is contained in $[0, U]$ and every target category b,

$$\left| \frac{m_a^R}{r m_{n,a}} - 1 \right| \leq \delta, \qquad \left| \frac{N_{ab}^R}{r N_{n,ab}} - 1 \right| \leq \delta. \tag{2}$$

The initial history is $(U, 0, 1)$-regular. By Lemma 2.4, conditional on any residual history, the unexposed row matchings are independent permanental matchings on their remaining submatrices.

An *admissible exploration* starts from a residual history and reveals one outgoing matching edge at a time. Before an edge is revealed, its source must be measurable with respect to the filtration generated by the slab data, the residual history, and all previously exposed edges. We also permit a fixed number of distinguished targets to be forbidden, with the next edge in a row containing such a target sampled conditionally on avoiding it. Lemma 2.6 shows that every actual cycle exploration and every root-avoiding exploration used below is admissible.

Consider an injective orbit segment $v_0, v_1, \ldots, v_\ell$, where $v_{j+1} = \sigma_n(v_j)$ for the actual exploration, and allow $v_\ell = v_0$ only at a terminal closing step. For an auxiliary exploration, v_{j+1} denotes the target selected under its conditional avoid-target law. Put

$$R_a(\ell) = \#\{0 \leq j < \ell : v_j \in H_{n,a}\},$$

and

$$R_{ab}(\ell) = \#\{0 \leq j < \ell : v_j \in H_{n,a},\ v_{j+1} \in H_{n,b}\}.$$

Whenever no ambiguity is possible we suppress the argument ℓ. The deterministic flow identity is

$$R_b - \sum_a R_{ab} = \mathbf{1}_{\{v_0 \in H_{n,b}\}} - \mathbf{1}_{\{v_\ell \in H_{n,b}\}}. \tag{3}$$

Lemma 2.8 (Scale-sensitive two-scale orbit balance)**.** *Fix master-partition endpoints $0 < S < U \leq T$, $\xi > 0$, and $0 \leq \delta < 1/4$. There are deterministic sequences $\chi_n \downarrow 0$ and $\pi_n \downarrow 0$, depending on the fixed partition, U, and ξ but not on the residual history, with the following property. On $\mathcal{G}_n$, uniformly over every (U, δ, ξ)-regular history and every admissible exploration, with conditional probability at least $1 - \pi_n$ the following assertions hold simultaneously at every defined prefix length $\ell \geq 1$.*

For every row a whose clock interval is contained in $[0, U]$ and every target category b,

$$\left| R_{ab}(\ell) - \frac{R_a(\ell)}{m_a^R} N_{ab}^R \right| \leq (C\beta + \chi_n) N_{ab}^R, \tag{4}$$

where C is absolute and $\beta = 2\kappa h$. Define

$$\Delta_n(U, \delta, h) = \delta + \beta + \chi_n + \varepsilon_n + e^{-U/2}.$$

Then, for every inner state b with $J_b \subset [0, A(S)]$,

$$\left| \frac{R_b(\ell)}{\ell p_{n,b}} - 1 \right| \le C_{\kappa,S} \frac{n}{\ell} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J}}}{\ell}. \tag{5}$$

Here $C_{\kappa,S}$ is independent of the mesh and of ℓ; the finite constant $C_{\mathcal{J}}$ may depend on the fixed master partition. In particular, for every fixed $\lambda > 0$, all prefixes with $\ell \ge \lambda n$ satisfy

$$\max_{b: J_b \subset [0, A(S)]} \left| \frac{R_b(\ell)}{\ell p_{n,b}} - 1 \right| \le C_{\kappa,S,\lambda} \Delta_n(U, \delta, h) + o(1).$$

Proof. On $\mathcal{G}_n$, every $m_{n,a}$ and every $N_{n,ab}$ is bounded below by a positive constant times n. If the history is regular, the same is true for m_a^R and N_{ab}^R, uniformly over the stated histories. Consequently, for each row below U, the residual target categories have proportions bounded below by a positive constant depending only on the fixed partition. Apply Lemmas 2.6 and 2.7 to the adaptive sequence of queried sources in that row, extending it predictably after the orbit segment ends until all ordinary targets in that row have been selected. In a row with a forbidden target, this means a complete order of the allowable targets. A union bound over the finitely many rows and categories gives deterministic sequences $\chi_n, \pi_n \downarrow 0$ for which (4) holds simultaneously for every prefix. If finitely many distinguished targets are forbidden, remove them from the corresponding category counts; the resulting $O(1)$ changes are absorbed into χ_n.

Write $p = p_n$, $P = P_n$, $q_a = R_a/\ell$, and

$$E_{ab} = R_{ab} - R_a P_{ab}.$$

For a row below U, regularity gives

$$\frac{N_{ab}^R}{m_a^R} = P_{ab}\{1 + O(\delta)\},$$

uniformly, and $R_a \le m_a^R$. Hence

$$|E_{ab}| \le C(\delta + \beta + \chi_n) N_{n,ab} \qquad (a \notin D(U)), \tag{6}$$

where $D(U)$ is the union of rows whose clock intervals lie in (U, ∞). For an outer row we use only injectivity of the explored path: $R_{ab} \le N_{ab}^R \le N_{n,ab}$ and $R_a P_{ab} \le m_a^R P_{ab} \le N_{n,ab}$. Thus

$$|E_{ab}| \le 2N_{n,ab} \qquad (a \in D(U)). \tag{7}$$

Let

$$G_b = \sum_{a \notin D(U)} E_{ab}, \qquad H_b = \sum_{a \in D(U)} E_{ab}.$$

Since $\sum_a N_{n,ab} = m_{n,b} = np_b$, equation (6) gives

$$\left\| \frac{G}{\ell p} \right\|_{L^2(p)} \le C \frac{n}{\ell} (\delta + \beta + \chi_n). \tag{8}$$

For the outer rows, Cauchy's inequality and the Hilbert–Schmidt block estimate on $\mathcal{G}_n$ give

$$\sum_b \frac{H_b^2}{m_{n,b}} \le 4 \sum_b \frac{(\sum_{a\in D(U)} N_{n,ab})^2}{m_{n,b}}$$
$$\le 4 m_{n,D(U)} \sum_{a\in D(U)} \sum_b \frac{m_{n,a} P_{n,ab}^2}{m_{n,b}} \le C m_{n,D(U)}. \tag{9}$$

Therefore, using the tail-mass estimate on $\mathcal{G}_n$,

$$\left\| \frac{H}{\ell p} \right\|_{L^2(p)} \le C \frac{n}{\ell} \left(e^{-U/2} + \varepsilon_n \right). \tag{10}$$

Let d_b denote the right-hand side of (3). It has at most two nonzero coordinates, each of absolute value at most one. Dividing (3) by ℓ gives

$$q - qP = e, \qquad e_b = \frac{d_b + G_b + H_b}{\ell}.$$

Since the partition is fixed,

$$\left\| \frac{e}{p} \right\|_{L^2(p)} \le C \frac{n}{\ell} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J}}}{\ell}. \tag{11}$$

Put $u_a = q_a/p_a - 1$. Because $pP = p$ and $\sum_a p_a u_a = 0$,

$$u - P^* u = e/p.$$

The spectral-gap event in $\mathcal{G}_n$ yields

$$\|u\|_{L^2(p)} \le (1 - q_0)^{-1} \left\| \frac{e}{p} \right\|_{L^2(p)}. \tag{12}$$

It remains to recover a coordinatewise estimate for an inner state b. The inner-column bound and stationarity imply

$$\sum_a p_a \left(\frac{P_{ab}}{p_b} \right)^2 \le \left(\max_a \frac{P_{ab}}{p_b} \right) \sum_a \frac{p_a P_{ab}}{p_b} \le C_{\kappa,S}.$$

Hence

$$|(P^* u)_b| \le C_{\kappa,S} \|u\|_{L^2(p)}.$$

For the coordinate $(e/p)_b$, equations (6) and (7), together with $P_{ab} \le C_{\kappa,S} p_b$ for an inner column, give

$$\left| \frac{e_b}{p_b} \right| \le C_{\kappa,S} \frac{n}{\ell} \left(\delta + \beta + \chi_n + e^{-U} + \varepsilon_n^2 \right) + \frac{C_{\mathcal{J}}}{\ell}.$$

Combining this estimate with $u_b = (e/p)_b + (P^* u)_b$, (11), and (12) proves (5). Every probabilistic input was the conditional relative-prefix event, so the failure probability π_n is uniform over the entire stated class of histories and explorations. □

2.5 The cycle of a uniform residual root

For a vertex v, let $c(v)$ be the unique row index for which $v \in Q_{n,c(v)}$. An orbit can hit v only when its current source lies in $H_{n,c(v)}$.

Lemma 2.9 (Root-avoiding change of measure)**.** *Fix a residual history and a distinguished remaining target v. Explore the path from v. At the jth opportunity to hit v, let λ_j be the conditional probability, under the original residual matching law and given that v has not yet been selected, that the next target is v. Let $\mathbb{P}^\circ$ be the auxiliary law obtained by conditioning each such choice not to select v. If $S(k)$ is the number of closure opportunities among the first k completed edges, then*

$$\mathbb{P}(|\mathcal{C}(v)| > k \mid \mathcal{H}, v) = \mathbb{E}^\circ \prod_{j=0}^{S(k)-1} (1-\lambda_j), \tag{13}$$

where the product is interpreted as zero if the auxiliary exploration is forced to stop before completing k edges. If the closure row initially has m available targets, including v, then at its jth closure opportunity,

$$\frac{e^{-\beta}}{m-j} \le \lambda_j \le \frac{e^{\beta}}{m-j}, \qquad \beta = 2\kappa h. \tag{14}$$

Proof. At a closure opportunity, the original transition law is the mixture of an atom of mass λ_j at v and the conditional avoid-v law of mass $1-\lambda_j$. A non-root previously visited vertex cannot be selected, because its target column has already been used; thus the only possible first repeat is the root. Multiplying the sequential Radon–Nikodym factors gives (13). Before the jth opportunity, exactly j non-v targets have been removed from the closure row, so $m-j$ targets remain. Equation (14) follows from hereditary flatness. If only v remains, the next closure probability is one, and the survival product is zero, which agrees with the stated convention. □

Lemma 2.10 (Harmonic-product approximation)**.** *Fix $\zeta \in (0,1)$. Uniformly for $c \in [1/2, 2]$, integers $m \ge 1$, and $0 \le s \le (1-\zeta)m$ for which all factors are nonnegative,*

$$\prod_{j=0}^{s-1} \left(1 - \frac{c}{m-j}\right) = (1-s/m)^c \left\{1 + O_\zeta(m^{-1})\right\}.$$

Moreover, uniformly for $x \in [\zeta, 1-\zeta]$ and $|a| \le 1/2$,

$$|(1-x)^{1+a} - (1-x)| \le C_\zeta |a|.$$

Proof. Since $m - j \ge \zeta m$,

$$\sum_{j<s} \log\left(1 - \frac{c}{m-j}\right) = -c \sum_{j<s} \frac{1}{m-j} + O_\zeta(m^{-1}).$$

The harmonic sum equals $\log(m/(m-s)) + O_\zeta(m^{-1})$, which proves the first assertion after exponentiation. The second follows from the mean-value theorem in the exponent and the boundedness of $|(1-x)^b \log(1-x)|$ for $x \in [\zeta, 1-\zeta]$ and $b \in [1/2, 3/2]$. □

Proposition 2.11 (Uniform conditional residual fraction)**.** *Fix master-partition endpoints $0 < S < U \le T$, $\xi > 0$, $\zeta \in (0, 1/2)$, and $0 \le \delta < 1/4$. On $\mathcal{G}_n$, let $\mathcal{H}$ be a (U, δ, ξ)-regular*

history. Choose a root uniformly from its N remaining vertices, independently of the unexposed matchings, and let L be the root-cycle length. With χ_n, π_n as in Lemma 2.8, uniformly over all such histories,

$$\sup_{x\in[\zeta,1-\zeta]} \left| \mathbb{P}\left(\frac{L}{N} \le x \,\middle|\, \mathcal{H} \right) - x \right| \\ \le C_{\kappa,S,\xi,\zeta}\big(\delta + \beta + \chi_n + \varepsilon_n + e^{-U/2}\big) + C_\xi(e^{-S} + \varepsilon_n^2) + \pi_n + \frac{C_{\mathcal{J},\xi,\zeta}}{n}. \tag{15}$$

The constants multiplying the principal error are independent of the mesh; $C_{\mathcal{J},\xi,\zeta}$ may depend on the fixed master partition.

Proof. Let v be the uniform root and put $C = c(v)$. Complete-cycle deletion gives $|Q_{n,a} \cap R| = m_a^R$, so, conditional on the history,

$$\mathbb{P}(C = a \mid \mathcal{H}) = m_a^R/N.$$

Consequently, using $N \ge \xi n$ and the tail bound on $\mathcal{G}_n$,

$$\mathbb{P}(C \in D(S) \mid \mathcal{H}) \le C_\xi(e^{-S} + \varepsilon_n^2). \tag{16}$$

We henceforth work on the event $C \notin D(S)$. Put $m = m_C^R$ and $p_C = p_{n,C}$. Regularity gives

$$\frac{m}{Np_C} = 1 + O(\delta). \tag{17}$$

For the fixed master partition, p_C is bounded below over the finitely many rows below S; thus $m \ge c_{\mathcal{J},S}\xi n$ for all large n.

We first rule out premature termination of the root-avoiding exploration. Let r be the number of completed edges if that exploration stops because only v remains in the closure row. Then exactly $m-1$ closure-row targets have been used, so

$$R_C(r) = m - 1, \qquad r \ge m - 1.$$

On the simultaneous prefix event of Lemma 2.8, its scale-sensitive estimate at this terminal prefix gives

$$\left| \frac{m-1}{rp_C} - 1 \right| \le C_{\kappa,S} \frac{n}{r} \Delta_n(U,\delta,h) + \frac{C_{\mathcal{J}}}{r}.$$

Writing $\theta = r/N$ and multiplying by θ yields

$$\left| \frac{m-1}{Np_C} - \theta \right| \le C_{\kappa,S,\xi} \Delta_n(U,\delta,h) + \frac{C_{\mathcal{J}}}{N}.$$

Together with (17),

$$|1 - r/N| \le C_{\kappa,S,\xi} \Delta_n(U,\delta,h) + \frac{C_{\mathcal{J},S,\xi}}{n}. \tag{18}$$

If the right-hand side is smaller than $\zeta/2$, the auxiliary exploration cannot stop before $(1-\zeta)N$ completed edges. If it is not smaller than $\zeta/2$, the right-hand side of (15), after increasing its constant, is at least one and the proposition is trivial. Thus we may assume that the auxiliary exploration is defined through every $k \le (1-\zeta)N$.

Fix $x \in [\zeta, 1-\zeta]$ and put $k = \lfloor xN \rfloor$. The number $S(k)$ of closure opportunities is exactly $R_C(k)$ under the avoid-v law. Applying (5) at $\ell = k$, and using $k \geq \zeta\xi n$, gives, on the simultaneous prefix event,

$$\left| \frac{S(k)}{kp_C} - 1 \right| \leq C_{\kappa,S,\xi,\zeta} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J},\xi,\zeta}}{n}.$$

Combining this with (17) and $k/N = x + O(n^{-1})$ gives

$$\sup_{x \in [\zeta, 1-\zeta]} \left| \frac{S(\lfloor xN \rfloor)}{m} - x \right| \leq C_{\kappa,S,\xi,\zeta} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J},\xi,\zeta}}{n}. \tag{19}$$

The exceptional conditional probability is at most π_n.

On the event in (19), the ratio $S(k)/m$ lies in $[\zeta/2, 1 - \zeta/2]$ once the displayed error is small; in the complementary parameter regime the final estimate is again trivial. Equations (13) and (14) squeeze the survival product between

$$\prod_{j=0}^{S(k)-1} \left(1 - \frac{e^{\beta}}{m-j}\right) \quad \text{and} \quad \prod_{j=0}^{S(k)-1} \left(1 - \frac{e^{-\beta}}{m-j}\right).$$

Lemma 2.10, followed by (19), shows that both products differ from $1 - x$ by at most

$$C_{\kappa,S,\xi,\zeta} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J},\xi,\zeta}}{n}.$$

On the exceptional prefix event we use the trivial bound by one. Averaging under $\mathbb{P}^{\circ}$, then adding the closure-tail probability (16), gives

$$\sup_{x \in [\zeta, 1-\zeta]} |\mathbb{P}(L > \lfloor xN \rfloor \mid \mathcal{H}) - (1 - x)|$$

bounded by the right-hand side of (15). Since $\{L/N \leq x\} = \{L \leq \lfloor xN \rfloor\}$, this is the claimed CDF estimate. □

2.6 Deletion stability and regeneration

Lemma 2.12 (Deletion of an interior macroscopic cycle)**.** *Fix master-partition endpoints $0 < V < U \leq T$, $\xi > 0$, $\zeta \in (0, 1/2)$, and $0 \leq \delta < 1/4$. On $\mathcal{G}_n$, start from a (U, δ, ξ)-regular history and expose the cycle of a uniform residual root. Let N and L be the residual size and the exposed cycle length. With χ_n, π_n as in Lemma 2.8, there is a constant $C = C(\kappa, V, \xi, \zeta)$ such that*

$$\mathbb{P}\Big(\zeta N \leq L \leq (1-\zeta)N, \text{ the post-deletion history is not } (V, \delta', \zeta\xi)\text{-regular} \;\Big|\; \mathcal{H}\Big) \leq \pi_n, \tag{20}$$

where

$$\delta' = C\big(\delta + \beta + \chi_n + \varepsilon_n + e^{-U/2}\big) + \frac{C_{\mathcal{J},\xi,\zeta}}{n}. \tag{21}$$

The conclusion is asserted whenever the displayed δ' is smaller than $1/4$, which is the only regime used below. This is a joint-event estimate; it does not condition on the exact random value of L.

Proof. Let d_a be the number of vertices of the exposed cycle in $H_{n,a}$, and let d_{ab} be the number of its directed edges from $H_{n,a}$ to $H_{n,b}$. The conditional prefix event in Lemma 2.8 is simultaneous over all prefix lengths. It may therefore be evaluated at the random closing time without conditioning on L.

On the event $\zeta N \le L \le (1-\zeta)N$, we have $L \ge \zeta\xi n$. Apply the scale-sensitive estimate (5) with inner cutoff V and $\ell = L$. For every row below V,

$$\left| \frac{d_a}{L p_{n,a}} - 1 \right| \le C_{\kappa,V,\xi,\zeta} \Delta_n(U,\delta,h) + \frac{C_{\mathcal{J},\xi,\zeta}}{n}.$$

Regularity gives $m_a^R = N p_{n,a}(1+O(\delta))$, and hence

$$\frac{d_a}{m_a^R} = \frac{L}{N} + O\left(\delta + \beta + \chi_n + \varepsilon_n + e^{-U/2} + \frac{C_{\mathcal{J},\xi,\zeta}}{n}\right) \tag{22}$$

uniformly over rows below V.

The row-prefix estimate (4), evaluated at the same random closing time, gives

$$d_{ab} = \frac{d_a}{m_a^R} N_{ab}^R + O((\beta+\chi_n) N_{ab}^R).$$

Combining this with (22),

$$d_{ab} = \frac{L}{N} N_{ab}^R + O\left(\delta + \beta + \chi_n + \varepsilon_n + e^{-U/2} + \frac{C_{\mathcal{J},\xi,\zeta}}{n}\right) N_{ab}^R. \tag{23}$$

The quantity d_{ab} is exactly the number of vertices removed from $Q_{n,a} \cap H_{n,b}$, because every cycle edge from row a to category b uses one such target.

Write $x = L/N$ and let R' be the post-deletion vertex set. Equations (22)–(23) give

$$m_a^{R'} = (1-x) m_a^R + O(\Delta_n^* m_a^R), \qquad N_{ab}^{R'} = (1-x) N_{ab}^R + O(\Delta_n^* N_{ab}^R),$$

where Δ_n^* denotes the parenthesized error in (22). Since $1 - x \ge \zeta$ and $|R'| = (1-x)N \ge \zeta\xi n$, division by the new total fraction preserves the regularity inequalities with error at most the quantity in (21). Residual permanental factorization and hereditary flatness remain valid after deleting the exposed edges. The only conditional failure event is the simultaneous prefix failure, whose probability is at most π_n. This proves (20). □

2.7 Independent residual fractions

Order the cycles size-biased by choosing, at each stage, a new root independently and uniformly from the current residual vertex set, deleting its cycle, and repeating. Let

$$V_{n,j} = \frac{\widetilde{L}_{n,j}}{n - \widetilde{L}_{n,1} - \cdots - \widetilde{L}_{n,j-1}}$$

whenever the denominator is positive, and set $V_{n,j} = 0$ after exhaustion.

Lemma 2.13 (Conditional test functions from an interior CDF bound)**.** *Let $W \in [0,1]$ and let $\mathscr{F}$ be a sigma-field. Suppose that, almost surely,*

$$\sup_{x \in [a,b]} |\mathbb{P}(W \le x \mid \mathscr{F}) - x| \le \eta, \qquad 0 < a < b < 1.$$

Then

$$\mathbb{P}(W \notin [a,b] \mid \mathscr{F}) \le a + (1-b) + 2\eta.$$

If f is absolutely continuous, vanishes outside $[a,b]$, and satisfies $f(a) = f(b) = 0$, then

$$\left| \mathbb{E}(f(W) \mid \mathscr{F}) - \int_0^1 f(u)\,du \right| \le \eta \int_a^b |f'(u)|\,du.$$

Proof. The endpoint estimate follows by evaluating the CDF bound at a and b. For the second assertion, conditional Stieltjes integration by parts and the vanishing boundary values give

$$\mathbb{E}(f(W) \mid \mathscr{F}) = -\int_a^b \mathbb{P}(W \le u \mid \mathscr{F}) f'(u)\,du,$$

whereas

$$\int_a^b f(u)\,du = -\int_a^b u f'(u)\,du.$$

Subtracting proves the claim. □

Proposition 2.14 (Finite-dimensional GEM(1) limits)**.** *For every fixed k,*

$$(V_{n,1}, \dots, V_{n,k}) \Rightarrow (U_1, \dots, U_k),$$

where $U_1, \dots, U_k$ are independent Unif$(0,1)$ *variables.*

Proof. Fix k and $\zeta \in (0, 1/4)$, and put $\xi_* = \zeta^{k-1}$. We first record a noncircular parameter hierarchy. Let $\eta > 0$ be a target error.

Choose T_0 so large that the closure-tail term $C_{\xi_*} e^{-T_0}$ in Proposition 2.11 is below a prescribed small fraction of η. Suppose $T_0, \dots, T_j$ have been chosen. Let R_j and D_j be, respectively, upper bounds for the principal-error coefficients in the root law and deletion lemma with inner cutoff T_j, residual fraction at least ξ_*, and truncation parameter ζ. These constants are now fixed. An error created at this level can subsequently be amplified only by $D_{j-1}, \dots, D_0$, all of which have already been fixed. Put

$$A_j = (1 + R_j + D_j) \prod_{i=0}^{j-1} (1 + D_i),$$

with the empty product equal to one, and choose $T_{j+1} > T_j$ so large that

$$A_j e^{-T_{j+1}/2} \le 2^{-j-3}\eta.$$

Continue until T_k has been chosen.

Use one master partition of $[0, T_k]$ containing $T_0, \dots, T_k$, append the tail interval, and let its mesh be at most h. After the cutoffs are fixed, all coefficients in the finitely many root and deletion estimates are finite. Choose h so small that the errors involving $\beta = 2\kappa h$ remain below their assigned fractions of η after the finite deletion recursion. More explicitly, if D_j denotes the coefficient in (21) with inner cutoff T_j, the deterministic recursion

$$d_0 = 0, \qquad d_{r+1} = D_{k-r-1}(d_r + \beta + e^{-T_{k-r}/2})$$

has $\max_{r \le k} d_r \to 0$ as all displayed input errors tend to zero. The preceding choices make this maximum smaller than a prescribed fraction of η, and in particular smaller than $1/8$. Finally,

for this fixed master partition take the maxima of the finitely many sequences χ_n and π_n that occur at the different cutoff levels, and choose n so large that

$$\varepsilon_n + \chi_n + \pi_n + C_{\mathcal{J}}/n$$

is below its assigned fraction of η and $\mathbb{P}(\mathcal{G}_n^c) \le \eta$. This order of choices is important: the inner cutoff is fixed before its outer cutoff, and the mesh is chosen only after all cutoff-dependent constants are known.

For reference, the cutoff used at each deletion stage is summarized by

stage	current regularity	root-law inner	root-law outer	post-deletion regularity
1	T_k	T_{k-1}	T_k	T_{k-1}
r	T_{k-r+1}	T_{k-r}	T_{k-r+1}	T_{k-r}
k	T_1	T_0	T_1	T_0

Thus every root-law inner cutoff was fixed before its corresponding outer cutoff, and deletion moves exactly one level inward in the already chosen hierarchy.

On the event that

$$\zeta \le V_{n,j} \le 1-\zeta, \qquad 1 \le j < r, \tag{6.20}$$

the residual size before stage r is at least $\zeta^{r-1}n \ge \xi_* n$. At stage r, use outer cutoff T_{k-r+1} and inner cutoff T_{k-r}. Inductively, the current history is regular at the outer cutoff with a regularity error bounded by the deterministic recursion above, except on a conditional event of probability $O_k(\eta)$. Proposition 2.11 therefore gives, uniformly over every such current history,

$$\sup_{x\in[\zeta,1-\zeta]} |\mathbb{P}(V_{n,r} \le x \mid \mathscr{F}_{r-1}) - x| \le C_k\eta. \tag{24}$$

On the event $V_{n,r} \in [\zeta, 1-\zeta]$, Lemma 2.12 makes the next history regular at cutoff T_{k-r}, except on a conditional event of probability at most $C_k\eta$. All these estimates are uniform over the preceding regular history.

We now turn the conditional estimates into joint convergence. Let $f_1, \dots, f_k$ be bounded Lipschitz functions on $[0,1]$. Choose a piecewise linear cutoff φ_ζ which is zero outside $[\zeta, 1-\zeta]$ and equal to one on $[2\zeta, 1-2\zeta]$, and put $g_j = f_j\varphi_\zeta$. Lemma 2.13, applied to (24), gives

$$\left|\mathbb{E}(g_r(V_{n,r}) \mid \mathscr{F}_{r-1}) - \int_0^1 g_r(u)\,du\right| \le C_{f_r,\zeta}\eta$$

on every regular history. Let $\mathcal{R}_r$ be the event that, after the first r deletions, the history is regular at cutoff T_{k-r} with the prescribed error; put $\mathcal{R}_0 = \mathcal{G}_n$. Because g_r vanishes outside $[\zeta, 1-\zeta]$, the joint-event deletion lemma also gives, on $\mathcal{R}_{r-1}$,

$$\left|\mathbb{E}\big(g_r(V_{n,r})\mathbf{1}_{\mathcal{R}_r} \mid \mathscr{F}_{r-1}\big) - \int_0^1 g_r(u)\,du\right| \le C_{f_r,\zeta}\eta.$$

Multiplying by $\prod_{j<r} g_j(V_{n,j})\mathbf{1}_{\mathcal{R}_{r-1}}$, taking expectations, and inducting on r yields

$$\left|\mathbb{E}\left(\prod_{j=1}^k g_j(V_{n,j})\mathbf{1}_{\mathcal{R}_k}\right) - \prod_{j=1}^k \int_0^1 g_j(u)\,du\right| \le C_{f,k,\zeta}\eta + o_n(1).$$

On the event that every $g_j(V_{n,j})$ is nonzero, every residual fraction is in the interior range, so a union bound over the conditional deletion failures shows that removing $\mathbf{1}_{\mathcal{R}_k}$ costs another $C_{f,k}\eta + o_n(1)$. Consequently

$$\left| \mathbb{E} \prod_{j=1}^{k} g_j(V_{n,j}) - \prod_{j=1}^{k} \int_0^1 g_j(u)\, du \right| \le C_{f,k,\zeta}\eta + o_n(1). \tag{25}$$

Here the $o_n(1)$ includes $\mathbb{P}(\mathcal{G}_n^c)$.

Evaluating (24) at 2ζ and $1-2\zeta$, and adding the regularity-failure probabilities, gives

$$\limsup_{n\to\infty} \mathbb{P}\left(V_{n,j} \notin [2\zeta, 1-2\zeta] \text{ for some } j \le k\right) \le 4k\zeta + C_k\eta. \tag{26}$$

Consequently, replacing g_j by f_j in (25) costs at most $C_{f,k}(\zeta+\eta)$, and the same is true for replacing $\int g_j$ by $\int f_j$. Let first $n\to\infty$, then $\eta \downarrow 0$, and finally $\zeta \downarrow 0$. We obtain

$$\mathbb{E} \prod_{j=1}^{k} f_j(V_{n,j}) \longrightarrow \prod_{j=1}^{k} \int_0^1 f_j(u)\, du.$$

Products of bounded Lipschitz functions form a convergence-determining class on $[0,1]^k$. This proves convergence to k independent uniform variables. The convention $V_{n,j} = 0$ after exhaustion causes no difficulty, because exhaustion before stage k is contained in the endpoint event in (26). □

Proof of Theorem 2.1. Put

$$X_{n,j} = \frac{\widetilde{L}_{n,j}}{n} = V_{n,j} \prod_{i<j} (1 - V_{n,i}), \qquad R_{n,k} = 1 - \sum_{j=1}^{k} X_{n,j} = \prod_{i=1}^{k} (1 - V_{n,i}).$$

Proposition 2.14 and the continuous mapping theorem give, for every fixed k,

$$\begin{aligned} &(X_{n,1}, \dots, X_{n,k}, R_{n,k}) \\ &\quad \Rightarrow \left(U_1, (1-U_1)U_2, \dots, U_k \prod_{i<k}(1-U_i), \prod_{i=1}^{k}(1-U_i) \right). \end{aligned} \tag{27}$$

Because $0 \le R_{n,k} \le 1$,

$$\mathbb{E}R_{n,k} \longrightarrow 2^{-k}.$$

Thus, for every $\epsilon > 0$,

$$\limsup_{n\to\infty} \mathbb{P}(R_{n,k} > \epsilon) \le \epsilon^{-1} 2^{-k}.$$

Equation (2.7) records the corresponding probabilistic tail bound, while the expectation identity will give the sharper bounded-Lipschitz truncation estimate below. Let $X_n = (X_{n,1}, X_{n,2}, \dots)$ and let X denote the GEM(1) residual-allocation sequence. Write $X_n^{(k)}$ and $X^{(k)}$ for truncation after the first k coordinates, with all later coordinates set to zero. If $F : \ell^1 \to \mathbb{R}$ is bounded and Lipschitz, then

$$\begin{aligned} |\mathbb{E}F(X_n) - \mathbb{E}F(X)| &\le \mathrm{Lip}(F)\big(\mathbb{E}R_{n,k} + \mathbb{E}R_k\big) \\ &\quad + |\mathbb{E}F(X_n^{(k)}) - \mathbb{E}F(X^{(k)})|, \end{aligned}$$

where $R_k = \sum_{j>k} X_j = \prod_{i=1}^{k}(1-U_i)$. For fixed k, the last term tends to zero by (27) and the continuous mapping theorem, while

$$\mathbb{E}R_{n,k} \longrightarrow 2^{-k} = \mathbb{E}R_k.$$

Taking first $n \to \infty$ and then $k \to \infty$ proves convergence against every bounded Lipschitz functional. By the bounded-Lipschitz criterion on the separable metric space ℓ^1,

$$(X_{n,1}, X_{n,2}, \ldots) \Rightarrow \mathrm{GEM}(1)$$

in ℓ^1.

For finite nonnegative sequences, the usual two-point uncrossing inequality shows that matching coordinates in decreasing order minimizes the sum of absolute differences. Passing to truncations gives

$$\|x^{\downarrow} - y^{\downarrow}\|_1 \le \|x - y\|_1$$

for nonnegative ℓ^1 sequences. The continuous mapping theorem therefore gives the PD(1) limit. The first residual fraction is the cycle length of a uniform root divided by n, which proves the final assertion of the theorem.

The argument was carried out along an arbitrary subsequence after extracting a marked-profile limit. The resulting law does not depend on that profile. Every subsequence therefore has a further subsequence with the same limit, so the full sequence converges. □

Corollary 2.15 (Power sums of macroscopic cycles)**.** *Under the hypotheses of Theorem 2.1, for every $q > 1$,*

$$\sum_{j\ge1}\left(\frac{L_{n,j}}{n}\right)^q \Rightarrow \sum_{j\ge1} P_j^q, \qquad (P_j) \sim \mathrm{PD}(1),$$

and

$$\mathbb{E}\sum_{j\ge1}\left(\frac{L_{n,j}}{n}\right)^q \longrightarrow \frac{1}{q}.$$

Proof. The map $x \mapsto \sum_j x_j^q$ is continuous and bounded by one on $\mathcal{S}^{\downarrow}$. If $\widetilde{P}$ is a size-biased pick from a partition (P_j), then, conditionally on the partition,

$$\mathbb{E}(\widetilde{P}^{\,q-1} \mid (P_j)) = \sum_j P_j^q.$$

For a PD(1) partition, its size-biased ordering has the GEM(1) law, so $\widetilde{P}$ is uniform on $(0,1)$. Hence

$$\mathbb{E}\sum_j P_j^q = \mathbb{E}\widetilde{P}^{\,q-1} = \int_0^1 u^{q-1}\,du = 1/q.$$

Weak convergence and bounded continuity of the power-sum functional transfer the expectation limit. □

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